

UNIVERSITY OF SOUTHERN CALIFORNIA

EE 483: Digital Signal Processing

Lecture 8: The Discrete Fourier Transform (DFT)

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1 The Discrete Fourier Transform (DFT)

Def.

The DFT is a mapping of a length- N sequence $x[n]$, $0 \leq n \leq N - 1$, into a length- N Fourier representation $X[k]$, $k = 0, \dots, N - 1$, according to

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi kn}{N}}. \quad (\text{Mitra 5.7})$$

It is a **sampled version of the DTFT** for a finite-length sequence:¹

$$X[k] = X(e^{j\omega}) \Big|_{\omega = \frac{2\pi k}{N}}.$$

$\Rightarrow$ discrete in both time and frequency.

Inverse DFT.

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{+j \frac{2\pi kn}{N}}, \quad n = 0, \dots, N - 1.$$

¹See the DTFT definition in Lecture 5 (September 8): $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$ (Mitra 3.10).

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1.1 Twiddle-Factor Notation

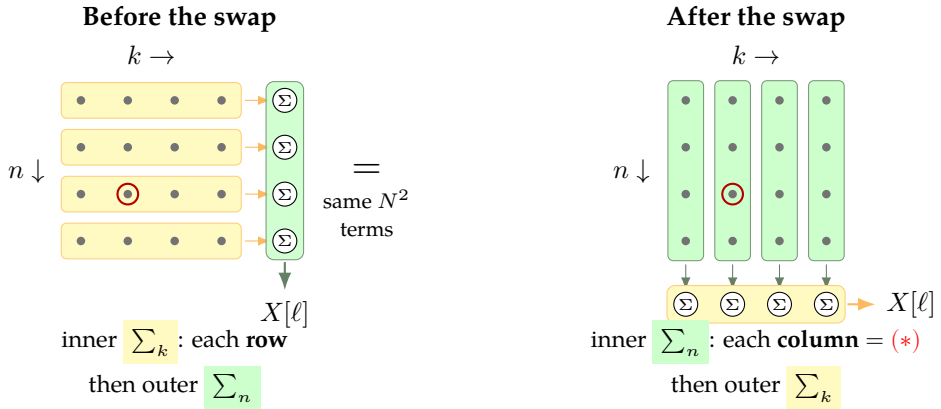
With $W_N = e^{-j\frac{2\pi}{N}}$,

Rewrite the DFT: $X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, \quad 0 \leq k \leq N-1. \quad (\text{Mittra 11.60})$

Rewrite the Inverse DFT: $x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}, \quad 0 \leq n \leq N-1.$

Verification of the inverse formula (check that the IDFT is indeed the inverse of the DFT).

$$\begin{aligned}
 X[\ell] &= \sum_{n=0}^{N-1} x[n] W_N^{\ell n} \\
 &= \sum_{n=0}^{N-1} \left[\frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn} \right] W_N^{\ell n} \\
 &= \sum_{n=0}^{N-1} \sum_{k=0}^{N-1} \frac{1}{N} X[k] W_N^{-kn} W_N^{\ell n} \quad (\text{move all factors into } \sum_k) \\
 &= \sum_{k=0}^{N-1} \sum_{n=0}^{N-1} \frac{1}{N} X[k] W_N^{-kn} W_N^{\ell n} \quad (\text{swap the sums}) \\
 &= \sum_{k=0}^{N-1} X[k] \underbrace{\left[\frac{1}{N} \sum_{n=0}^{N-1} W_N^{-(k-\ell)n} \right]}_{(*)}.
 \end{aligned}$$



Each dot is one term $a[n, k] = \frac{1}{N} X[k] W_N^{-kn} W_N^{\ell n}$ (drawn for $N = 4$; illustration, not in orig lecture note). The red ring is the same term $a[2, 1]$ in both grids: it moves from “row 2” to “column 1” but is not changed.

Summing the geometric series (recall $W_N = e^{-j\frac{2\pi}{N}}$, so $W_N^{-(k-\ell)} = e^{j\frac{2\pi(k-\ell)}{N}}$),

$$\begin{aligned}
 (*) &= \frac{1}{N} \frac{1 - W_N^{-(k-\ell)N}}{1 - W_N^{-(k-\ell)}} = \frac{1}{N} \frac{1 - e^{j2\pi(k-\ell)}}{1 - e^{j\frac{2\pi(k-\ell)}{N}}} \\
 &= \frac{1}{N} \frac{e^{j\pi(k-\ell)} (e^{-j\pi(k-\ell)} - e^{j\pi(k-\ell)})}{e^{j\frac{\pi(k-\ell)}{N}} (e^{-j\frac{\pi(k-\ell)}{N}} - e^{j\frac{\pi(k-\ell)}{N}})} \\
 &= \frac{1}{N} \frac{\sin(\pi[k - \ell]) e^{j\pi(k-\ell)}}{\sin(\frac{\pi}{N}[k - \ell]) e^{j\frac{\pi(k-\ell)}{N}}} \quad 2
 \end{aligned}$$

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$$= \begin{cases} 1, & k \equiv \ell \pmod{N}, \\ 0, & \text{else} \end{cases} \quad 3$$

²The $\sin / \sin$ factor is the *Dirichlet kernel* $D_N(\omega) = \frac{\sin(\omega \frac{N}{2})}{\sin(\frac{\omega}{2})}$ (Lecture 5, September 8, § 2), at $\omega = \frac{2\pi(\mathbf{k} - \ell)}{N}$.

In this verification, $k, \ell \in \{0, \dots, N-1\}$, so $k \equiv \ell \pmod{N} \iff k = \ell$.

$$= \delta[\langle k - \ell \rangle_N].$$

Modulo Operation $\langle n \rangle_N$.

$k = \langle n \rangle_N$ if $k \in \{0, \dots, N-1\}$ and there exists an integer ℓ such that $n = k + \ell N$.

Example $\langle n \rangle_3$ (see Mitra 2.3.1).

n	-2	-1	0	1	2	3	4	5	6
$\langle n \rangle_3$	1	2	0	1	2	0	1	2	0

Therefore

$$X[\ell] = \sum_{k=0}^{N-1} X[k] \delta[\langle k - \ell \rangle_N] = X[\ell] \quad \text{for } \ell \in \{0, \dots, N-1\}. \blacksquare$$

³The geometric-series formula is only valid when $W_N^{-(k-\ell)} \neq 1$, i.e., $k \not\equiv \ell \pmod{N}$; then $\sin(\pi[k - \ell]) = 0$ while $\sin(\frac{\pi}{N}[k - \ell]) \neq 0$, so $(*) = 0$. When $k \equiv \ell \pmod{N}$, $W_N^{-(k-\ell)} = 1$, so every term $W_N^{-(k-\ell)n} = 1$ and $(*) = \frac{1}{N} \cdot N = 1$ directly (also the 0/0 limit of the Dirichlet-kernel expression).

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Example.

$g[n] = \cos\left(\frac{2\pi rn}{N}\right)$, $0 \leq n \leq N-1$, with r an integer, $0 \leq r \leq N-1$. We can write $g[n]$ as

$$\begin{aligned} g[n] &= \frac{1}{2} \left(e^{j\frac{2\pi rn}{N}} + e^{-j\frac{2\pi rn}{N}} \right) && \text{(Euler)} \\ &= \frac{1}{2} (W_N^{-rn} + W_N^{rn}) && \text{(recall } W_N = e^{-j\frac{2\pi}{N}}). \end{aligned}$$

$$\begin{aligned} G[k] &= \sum_{n=0}^{N-1} g[n] W_N^{kn} = \sum_{n=0}^{N-1} \frac{1}{2} (W_N^{-rn} + W_N^{rn}) W_N^{kn} \\ &= \frac{1}{2} \left(\sum_{n=0}^{N-1} W_N^{-rn} W_N^{kn} + \sum_{n=0}^{N-1} W_N^{rn} W_N^{kn} \right) \\ &= \frac{1}{2} \left(\sum_{n=0}^{N-1} W_N^{-(r-k)n} + \sum_{n=0}^{N-1} W_N^{-(r+k)n} \right) \\ &= \frac{1}{2} \left(\textcolor{red}{N} \delta[\langle r-k \rangle_N] + \textcolor{red}{N} \delta[\langle -r-k \rangle_N] \right). \end{aligned}$$

For $0 \leq k \leq N-1$:

$$G[k] = \begin{cases} \frac{N}{2}, & k = r, \quad k = N-r, \\ 0, & \text{else,} \end{cases} \quad \text{(unless } r = 0 \text{ or } r = N/2)^4$$

using $\langle -r-k \rangle_N = \langle N-r-k \rangle_N$. When $r = 0$ or $r = N/2$, the two impulses coincide and $G[r] = N$.

Reading the two δ 's for $k \in \{0, \dots, N-1\}$:

- $k = r$: $\langle r-k \rangle_N = 0$ in the first term.
- $k = N-r$: $\langle -r-k \rangle_N = 0$ in the second term, i.e. $k \equiv -r \pmod{N}$ (the representative in $\{0, \dots, N-1\}$ is $N-r$).

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2 DFT Properties

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⁵cf. Mitra Tables 5.1, 5.2, and 5.3 (reproduced in the appendix of the full build).

2.1 Periodicity

Property.

$$\begin{aligned} X[\ell N + k] &= X[k], & k = 0, \dots, N-1, \ell \text{ an integer,} \\ x[mN + n] &= x[n], & n = 0, \dots, N-1, m \text{ an integer,} \end{aligned}$$

when the DFT / IDFT formulas are evaluated outside $0, \dots, N-1$.

Because

$$\begin{aligned} W_N^N &= e^{-j\frac{2\pi}{N}N} = e^{-j2\pi} = 1 \\ \Rightarrow W_N^{\ell N + k} &= (1)^\ell W_N^k = W_N^k. \end{aligned}$$

So adding ℓN to a frequency index (or mN to a time index) does not change any twiddle factor in the DFT/IDFT sums. We therefore extend $X[k]$ and $x[n]$ periodically with period N when those formulas are used for integer indices outside $\{0, \dots, N-1\}$ (as stated in the Property box).

2.2 Circular Shift

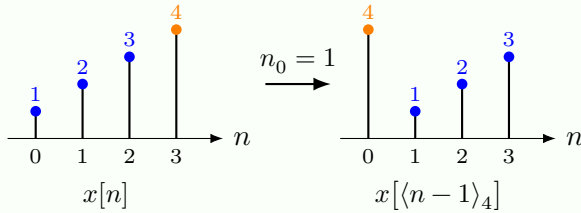
Consider $x[n]$, $0 \leq n \leq N-1$. The ordinary shift $x_1[n] = x[n - n_0]$ is **no longer defined** over $0 \leq n \leq N-1$.

$$\Rightarrow \text{circular shifts:} \quad x_1[n] = x[\langle n - n_0 \rangle_N]. \quad (\text{Mittra 2.25})$$

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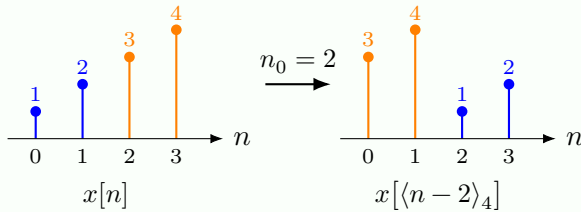
Example (Right Circular Shift).Let $0 < n_0 \leq N - 1$:

$$x_1[n] = x[\langle n - n_0 \rangle_N] = \begin{cases} x[n - n_0], & n_0 \leq n \leq N - 1, \\ x[N + n - n_0], & 0 \leq n \leq n_0 - 1. \end{cases} \quad (\text{Mitra 2.26})$$



$$x[n] = \{\underset{\uparrow}{1}, 2, 3, 4\}, \quad x[\langle n-1 \rangle_4] = \{\underset{\uparrow}{4}, 1, 2, 3\},$$

$$x[\langle n-2 \rangle_4] = \{\underset{\uparrow}{3}, 4, 1, 2\} = x[\langle n+2 \rangle_4].$$

**Property (Circular Time-Shifting).**Let $g[n] \leftrightarrow G[k]$ and $\tilde{g}[n] = g[\langle n - n_0 \rangle_N]$. Then

$$\tilde{G}[k] = W_N^{kn_0} G[k] = e^{-j\frac{2\pi kn_0}{N}} G[k].$$

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Proof.

$$\tilde{G}[k] = \sum_{n=0}^{N-1} g[\langle n - n_0 \rangle_N] W_N^{kn} = \sum_{n=0}^{n_0-1} g[n + N - n_0] W_N^{kn} + \sum_{n=n_0}^{N-1} g[n - n_0] W_N^{kn}.$$

Let $m = n + N - n_0$ ($n = m - N + n_0$) in the first sum and $p = n - n_0$ ($n = p + n_0$) in the second:

$$\begin{aligned} \tilde{G}[k] &= \sum_{m=N-n_0}^{N-1} g[m] W_N^{k(m-N+n_0)} + \sum_{p=0}^{N-1-n_0} g[p] W_N^{k(p+n_0)} \\ &= \sum_{n=0}^{N-1} g[n] W_N^{k(n+n_0)} \quad (W_N^{-kN} = 1) \\ &= W_N^{kn_0} \left[\sum_{n=0}^{N-1} g[n] W_N^{kn} \right] = W_N^{kn_0} G[k] = e^{-j\frac{2\pi kn_0}{N}} G[k]. \quad \blacksquare \end{aligned}$$

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2.3 Convolution

2.3.1 Linear convolution and the DTFT

$$x[n] \otimes h[n] \xleftrightarrow{\text{DTFT}} X(e^{j\omega}) H(e^{j\omega}).$$

If $x[n]$ has length N_1 and $h[n]$ has length N_2 , then $y[n] = x[n] \otimes h[n]$ has length $N_1 + N_2 - 1$: it is possibly nonzero for $0 \leq n \leq N_1 + N_2 - 2$ and is zero otherwise.

2.3.2 Circular Convolution (two length- N sequences)

Def.

$$y_c[n] = \sum_{m=0}^{N-1} x[m] h[\langle n - m \rangle_N], \quad 0 \leq n \leq N-1, \quad (\text{Mittra 5.51})$$

$$y_c[n] = x[n] \circledcirc h[n] = h[n] \circledcirc x[n].^6 \quad (\text{Mittra 5.52})$$

⁶The proof of Commutativity of Circular Convolution was asked in Problem Set 5, Problem 2.

Convolution Theorem.

If $y_c[n] = x[n] \circledast h[n]$, then $Y_c[k] = X[k] H[k]$.

9**Proof.**

$$\begin{aligned}
 Y_c[k] &= \sum_{n=0}^{N-1} \left[\sum_{m=0}^{N-1} x[\langle n-m \rangle_N] h[m] \right] W_N^{kn} \\
 &= \sum_{n=0}^{N-1} \sum_{m=0}^{N-1} x[\langle n-m \rangle_N] h[m] W_N^{kn} \quad (\text{expand inner sum}) \\
 &= \sum_{m=0}^{N-1} \sum_{n=0}^{N-1} x[\langle n-m \rangle_N] h[m] W_N^{kn} \quad (\text{swap the sums}) \\
 &= \sum_{m=0}^{N-1} h[m] \left[\sum_{n=0}^{N-1} x[\langle n-m \rangle_N] W_N^{kn} \right] \\
 &= \left[\sum_{m=0}^{N-1} h[m] W_N^{km} \right] X[k] \quad (\text{circular time-shift}) \\
 &= H[k] X[k].
 \end{aligned}$$

■

2.4 Linear Convolution via Circular Convolution

If $x[n], 0 \leq n \leq N_1 - 1$, and $h[n], 0 \leq n \leq N_2 - 1$, then

$$y[n] = x[n] \otimes h[n], \quad 0 \leq n \leq N_1 + N_2 - 2.$$

Property.

If

$$\tilde{x}[n] = \begin{cases} x[n], & 0 \leq n \leq N_1 - 1, \\ 0, & N_1 \leq n < N, \end{cases} \quad \text{where } N \geq N_1 + N_2 - 1,$$

and similarly define a “zero-padded” length- N $\tilde{h}[n]$, then

$$\tilde{x}[n] \textcircled{N} \tilde{h}[n] = x[n] \otimes h[n] \quad \text{for } 0 \leq n \leq N - 1.$$

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Consequence: we can compute $\otimes$ using $\textcircled{N}!$.

1. Zero-pad both sequences to the same length $N \geq N_1 + N_2 - 1$.
2. Take length- N DFTs of both.
3. Multiply the DFTs together.
4. Take the inverse DFT of the result.

Remark*(addendum to orig lecture note).*

Why $N \geq N_1 + N_2 - 1$: circular convolution equals the linear convolution *aliased* with period N , $\tilde{x}[n] \textcircled{N} \tilde{h}[n] = \sum_r y[n + rN]$ for $0 \leq n \leq N - 1$. Since $y[n]$ occupies $N_1 + N_2 - 1$ samples, no shifted copies overlap exactly when $N \geq N_1 + N_2 - 1$. With a fast algorithm (FFT) for the DFT, this is how long convolutions are computed in practice (cf. Mitra Ch. 11).

2.5 Parseval's Theorem**Property**

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$$\begin{aligned} \sum_{n=0}^{N-1} x^*[n] y[n] &= \frac{1}{N} \sum_{k=0}^{N-1} X^*[k] Y[k], \\ \Rightarrow \sum_{n=0}^{N-1} |x[n]|^2 &= \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2. \end{aligned} \quad (\text{Mitra 5.6})$$

The proof of Parseval's Theorem for the DFT was asked in Problem Set 5, Problem 1.

2.6 Symmetry

Property.

If $x[n]$ is real,

$$X[k] = X^*[N - k] = X^*[\langle -k \rangle_N], \quad 0 \leq k \leq N - 1.$$

(For $k = 0$, read $X^*[N]$ as $X^*[0]$ by periodicity; the modulo form handles this automatically.)

Readings

- Sanjit K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., Ch. 5.4, 5.5, 5.7, 11.3–11.5
- Monson H. Hayes, *Schaum's Outline of Digital Signal Processing*, 2nd ed., Ch. 6, 7

⁷Notation vs. Mitra:

- Sec. 5.1 is for a general orthogonal transform with basis $\psi[k, n]$. Its coefficients are written $\mathcal{X}[k]$ (calligraphic). $X[k]$ is used only for the DFT, from Sec. 5.2 on.
- Sec. 5.1 has only one sequence. Mitra (5.6) is the energy form $\sum_n |x[n]|^2 = \frac{1}{N} \sum_k |\mathcal{X}[k]|^2$.
- Its derivation puts the conjugate on the second factor: $\sum_n x[n] x^*[n] = \frac{1}{N} \sum_k \mathcal{X}[k] \mathcal{X}^*[k]$.
- The two-sequence line above is Mitra (5.115) in Sec. 5.7: $\sum_n g[n] h^*[n] = \frac{1}{N} \sum_k G[k] H^*[k]$.
- This lecture writes x, y instead of g, h and puts the conjugate on the first factor. Both sides are then the complex conjugates of Mitra's, so the equation still holds.